

COVID-19 Data Analysis

Testing Rates and Positive Cases: January–June 2020

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1 Introduction

This report analyses data on COVID-19 testing rates and positive test results between 20 January 2020 and 1 June 2020. The data source is provided by Dataquest, derived from Kaggle:

<https://dq-content.s3.amazonaws.com/505/covid19.csv>

Research question: Which countries have reported the highest number of positive cases in relation to the number of tests conducted?

2 Methods and Limitations

2.1 Scope and Sample

The analysis is restricted to the top ten countries by total testing volume. This is not a random or representative sample. It excludes countries with high death tolls but low testing capacity (e.g. Brazil, Mexico), which limits the generalisability of conclusions about which countries were most affected globally.

2.2 Positive-to-Tested Ratio

A high ratio of positive to tested cases does not straightforwardly indicate a higher infection rate. It more likely reflects a narrower testing strategy. For example, restricting tests to symptomatic or hospitalised patients. Comparisons of this ratio across countries should be interpreted as reflecting testing policy as much as disease burden.

2.3 Population Adjustment

Raw counts are not directly comparable across countries with very different population sizes. Sections 7–9 adjust for population by expressing metrics as a percentage of each country’s population. Population figures were provided by Dataquest and should be treated as estimates; no source date is given.

2.4 Missing Data

Several countries, including the US and UK, reported zero for **active** and **hospitalizedCurr**. These are non-reporting zeros rather than true zeros. This affects the aggregated impact rankings in section 9, where countries that reported all metrics will appear artificially more affected than those that did not.

2.5 Aggregated Rankings

Summing ranks across metrics (section 9) treats each metric as equally important and of comparable scale, which is not statistically justified. Results should be interpreted as a rough ordinal guide rather than a precise impact score. Countries with complete reporting will score higher regardless of true disease burden.

3 Understanding the Data

3.1 Setup

```
library(tidyverse)
library(skimr)
library(readr)
library(ggthemes)
library(zoo)
```

3.2 Loading the Dataset

```
url <- "https://dq-content.s3.amazonaws.com/505/covid19.csv"
covid_df <- read_csv(url)
# make two letter code and country_region variables consistent
covid_df <- covid_df %>%
  mutate(Two_Letter_Country_Code = recode(Two_Letter_Country_Code, "GB" =
    "UK"))
```

3.3 Initial Exploration

The following steps examine the dimensions, structure, and data types of the dataset, and flag any quality issues.

```
dim(covid_df)

[1] 10903    14

skim(covid_df)
```

Table 1: Data summary

Name	covid_df
Number of rows	10903
Number of columns	14

Column type frequency:

character	5
numeric	9

Group variables	None
-----------------	------

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Date	0	1	10	10	0	133	0
Continent_Name	0	1	4	13	0	6	0
Two_Letter_Country_Code	0	1	2	2	0	109	0
Country_Region	0	1	4	22	0	109	0
Province_State	0	1	4	28	0	81	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
positive	0	1	17768.02	93143.46	0	44.0	1026	7440.0	1783570	
hospitalized	0	1	766.40	5626.33	0	0.0	0	0.0	89590	
recovered	0	1	2409.22	11254.21	0	0.0	0	500.5	171883	
death	0	1	947.59	5507.24	0	0.0	6	136.0	98536	
total_tested	0	1	195475.72	19022.18	0	2147.5	30358	125285.01	6936891	
active	0	1	4472.19	22277.48	-2	0.0	0	660.5	280931	
hospitalizedCurr	0	1	454.07	2253.05	0	0.0	0	28.0	33004	
daily_tested	0	1	6841.40	27198.46	-	71.0	1048	4688.5	492276	
daily_positive	0	1	497.88	2354.83	-4735	1.0	29	264.0	63047	

```
vector_cols <- colnames(covid_df)
```

```
head(covid_df)
```

	Date	Continent_Name	Two_Letter_Country_Code	Country_Region
1	2020-01-20	Asia	KR	South Korea
2	2020-01-22	North America	US	United States
3	2020-01-22	North America	US	United States
4	2020-01-23	North America	US	United States
5	2020-01-23	North America	US	United States
6	2020-01-24	Asia	KR	South Korea

	Province_State	positive	hospitalized	recovered	death	total_tested	active
1	All States	1	0	0	0	4	0
2	All States	1	0	0	0	1	0
3	Washington	1	0	0	0	1	0
4	All States	1	0	0	0	1	0
5	Washington	1	0	0	0	1	0
6	All States	2	0	0	0	27	0

	hospitalizedCurr	daily_tested	daily_positive
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0
6	0	5	0

3.4 Issues Identified

Review of the raw data surfaces the following issues requiring attention:

- **Date format:** date is stored as a character string and should be converted to a date type for time-series analysis.
- **Mixed geographic granularity:** Province_State conflates US states with country-level entries (e.g. “All States”, “China”) in the same column.
- **Inconsistent naming:** hospitalizedCurr deviates from the underscore convention used by other columns.
- **Negative values:** Some rows contain negative values for daily_tested, daily_positive, and active, indicating a data quality issue. These rows should be removed.

4 Data Cleaning and Preparation

4.1 Isolating National-Level Rows

Rows are filtered to retain only entries where `Province_State == "All States"`, giving a consistent national-level view. The `Province_State` column is then dropped as it is no longer informative.

```
covid_df_all_states <- covid_df %>%  
  filter(Province_State == "All States") %>%  
  select(-Province_State)  
  
head(covid_df_all_states)
```

	Date	Continent_Name	Two_Letter_Country_Code	Country_Region	positive
1	2020-01-20	Asia	KR	South Korea	1
2	2020-01-22	North America	US	United States	1
3	2020-01-23	North America	US	United States	1
4	2020-01-24	Asia	KR	South Korea	2
5	2020-01-24	North America	US	United States	1
6	2020-01-25	Oceania	AU	Australia	4

	hospitalized	recovered	death	total_tested	active	hospitalizedCurr
1	0	0	0	4	0	0
2	0	0	0	1	0	0
3	0	0	0	1	0	0
4	0	0	0	27	0	0
5	0	0	0	1	0	0
6	0	0	0	0	0	0

	daily_tested	daily_positive
1	0	0
2	0	0
3	0	0
4	5	0
5	0	0
6	0	0

4.2 Selecting Relevant Columns

Daily metrics are separated from cumulative metrics to allow focused analysis.

```
daily_cols <- c(
```

```

    "Date", "Country_Region", "active",
    "hospitalizedCurr", "daily_tested", "daily_positive"
)

```

```

covid_df_all_states_daily <- covid_df_all_states %>%
  select(all_of(daily_cols))

```

```

covid_df_all_states_cumulative <- covid_df_all_states %>%
  select(-all_of(daily_cols))

```

5 Identifying Countries with the Highest Number of Deaths

```

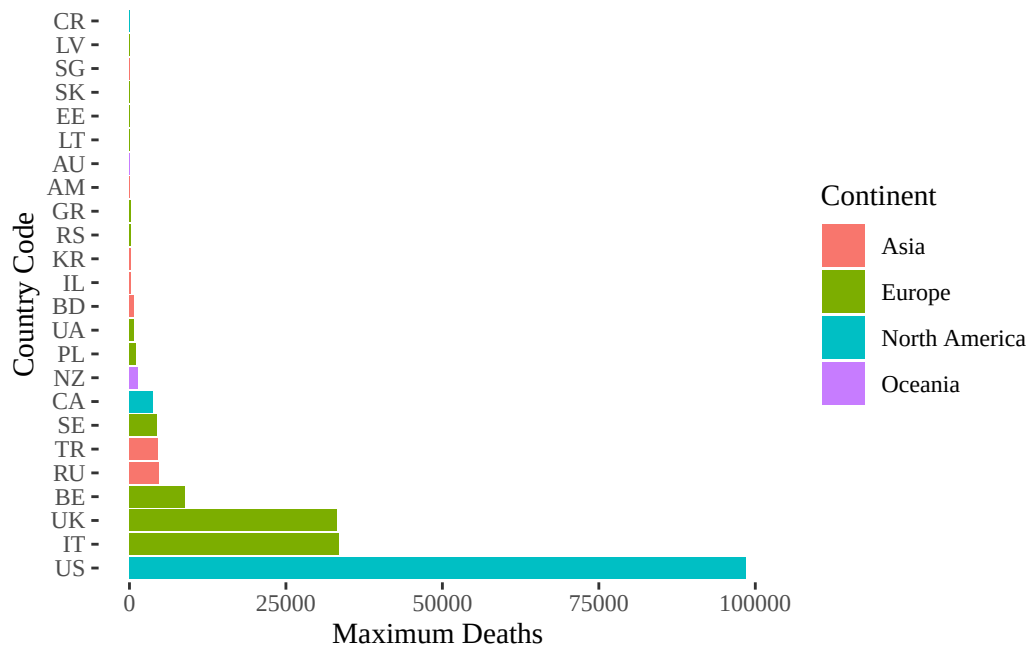
covid_df_all_states_cumulative_max <- covid_df_all_states_cumulative %>%
  group_by(Continent_Name, Two_Letter_Country_Code) %>%
  summarize(max_deaths = max(death)) %>%
  filter(max_deaths > 0)

```

```

ggplot(covid_df_all_states_cumulative_max,
       aes(x = reorder(Two_Letter_Country_Code, -max_deaths),
           y = max_deaths,
           fill = Continent_Name)) +
  geom_col() +
  labs(x = "Country Code", y = "Maximum Deaths", fill = "Continent") +
  theme_tufte() +
  coord_flip()

```

```
death_top_3 <- covid_df_all_states_cumulative_max %>%
  ungroup() %>%
  slice_max(max_deaths, n = 3) %>%
  pull(Two_Letter_Country_Code)
```

The US had, by far, the greatest maximum deaths, followed by Italy and Great Britain. The continents are affected in the following order: North America, Europe, Asia, Oceania. These are absolute values, not normalised to the respective countries' total populations.

6 Identifying the Top Ten Countries by Testing Volume

Countries are ranked by total tests conducted over the period. The top ten are extracted for further analysis.

```
covid_df_all_states_daily_sum <- covid_df_all_states_daily %>%
  group_by(Country_Region) %>%
  summarize(
    tested      = sum(daily_tested),
    positive    = sum(daily_positive),
    active      = sum(active),
    hospitalized = sum(hospitalizedCurr)
  ) %>%
  arrange(desc(tested))

covid_top_10 <- head(covid_df_all_states_daily_sum, 10)
```

```
covid_top_10
```

```
# A tibble: 10 x 5
```

	Country_Region	tested	positive	active	hospitalized
	<chr>	<int>	<int>	<int>	<int>
1	United States	17282363	1877179	0	0
2	Russia	10542266	406368	6924890	0
3	Italy	4091291	251710	6202214	1699003
4	India	3692851	60959	0	0
5	Turkey	2031192	163941	2980960	0
6	Canada	1654779	90873	56454	0
7	United Kingdom	1473672	166909	0	0
8	Australia	1252900	7200	134586	6655
9	Peru	976790	59497	0	0
10	Poland	928256	23987	538203	0

7 Positive-to-Tested Ratio Analysis

7.1 Computing the Ratio

The ratio of positive cases to tests conducted is calculated for each of the top ten countries to identify which had the highest relative burden.

```
countries      <- covid_top_10$Country_Region
tested_cases   <- setNames(covid_top_10$tested,      countries)
positive_cases <- setNames(covid_top_10$positive,    countries)
active_cases   <- setNames(covid_top_10$active,     countries)
hospitalized_cases <- setNames(covid_top_10$hospitalized, countries)
```

```
# Compute ratio and store as named vector for use in answer_list
```

```
positive_tested_top_3 <- covid_top_10 %>%
  mutate(positive_tested_ratio = positive / tested) %>%
  slice_max(positive_tested_ratio, n = 3) %>%
  select(Country_Region, positive_tested_ratio) %>%
  deframe()
```

```
# Display as formatted table
```

```
positive_tested_top_3 %>%
  as_tibble(rownames = "Country") %>%
```

```
knitr::kable(digits = 3, col.names = c("Country", "Positive / Tested Ratio"))
```

Country	Positive / Tested Ratio
United Kingdom	0.113
United States	0.109
Turkey	0.081

7.2 Summary for Top Three Countries

Key statistics for the three countries with the highest positive-to-tested ratio are collated below.

```
united_kingdom <- c(0.11, 1473672, 166909, 0, 0)
united_states  <- c(0.10, 17282363, 1877179, 0, 0)
turkey         <- c(0.08, 2031192, 163941, 2980960, 0)
```

```
covid_mat_top3 <- rbind(united_kingdom, united_states, turkey)
colnames(covid_mat_top3) <- c("Ratio", "Tested", "Positive", "Active",
"Hospitalized")
```

```
covid_mat_top3
```

	Ratio	Tested	Positive	Active	Hospitalized
united_kingdom	0.11	1473672	166909	0	0
united_states	0.10	17282363	1877179	0	0
turkey	0.08	2031192	163941	2980960	0

8 Scaling of Data to Population Level

```
covid_mat <- cbind(tested_cases, positive_cases, active_cases,
hospitalized_cases)
```

```
population <- c(31002651, 145934462, 60461826, 1380004385, 84339067,
37742154, 67886011, 25499884, 32971854, 37846611)
```

```
covid_mat <- covid_mat * 100 / population
```

```
covid_mat
```

	tested_cases	positive_cases	active_cases	hospitalized_cases
United States	55.744791	6.054898338	0.0000000	0.00000000
Russia	7.223973	0.278459244	4.7452054	0.00000000

Italy	6.766734	0.416312269	10.2580660	2.81004249
India	0.267597	0.004417305	0.0000000	0.00000000
Turkey	2.408364	0.194383227	3.5344949	0.00000000
Canada	4.384432	0.240773221	0.1495781	0.00000000
United Kingdom	2.170804	0.245866560	0.0000000	0.00000000
Australia	4.913356	0.028235423	0.5277906	0.02609816
Peru	2.962496	0.180447845	0.0000000	0.00000000
Poland	2.452679	0.063379519	1.4220639	0.00000000

9 Ranking Countries Related to Their Population

```

tested_cases_rank      <- rank(covid_mat[, "tested_cases"])
positive_cases_rank    <- rank(covid_mat[, "positive_cases"])
active_cases_rank      <- rank(covid_mat[, "active_cases"])
hospitalized_cases_rank <- rank(covid_mat[, "hospitalized_cases"])

```

```

covid_mat_rank <- cbind(
  tested_cases_rank, positive_cases_rank,
  active_cases_rank, hospitalized_cases_rank
)
covid_mat_rank

```

	tested_cases_rank	positive_cases_rank	active_cases_rank
United States	10	10	2.5
Russia	9	8	9.0
Italy	8	9	10.0
India	1	1	2.5
Turkey	3	5	8.0
Canada	6	6	5.0
United Kingdom	2	7	2.5
Australia	7	2	6.0
Peru	5	4	2.5
Poland	4	3	7.0
	hospitalized_cases_rank		
United States	4.5		
Russia	4.5		
Italy	10.0		
India	4.5		

Turkey	4.5
Canada	4.5
United Kingdom	4.5
Australia	9.0
Peru	4.5
Poland	4.5

10 Identifying Affected Countries Related to Their Population

Question 1: Which countries have made the best effort in terms of the number of tests conducted related to their population?

Question 2: Which countries were ultimately the most and least affected related to their population?

```
# Remove tested column; sum remaining metrics per country
covid_mat_rank_impact <- covid_mat_rank[, -1]
aggregated_rank <- rowSums(covid_mat_rank_impact)
aggregated_rank
```

United States	Russia	Italy	India	Turkey
17.0	21.5	29.0	8.0	17.5
Canada	United Kingdom	Australia	Peru	Poland
15.5	14.0	17.0	11.0	14.5

```
# Q1: top 3 testers per capita - highest rank = most tests relative to
population
best_effort_tested_cased_top_3 <- names(sort(tested_cases_rank, decreasing =
TRUE)[1:3])
```

```
# Q2: most and least affected by aggregated impact score
most_affected_country <- names(which.max(aggregated_rank))
least_affected_country <- names(which.min(aggregated_rank))
```

11 Time Series Analysis

```
# 7-day rolling average to smooth noise
```

```
Sys.setlocale("LC_TIME", "English")
```

```
[1] "English_United States.1252"
```

```

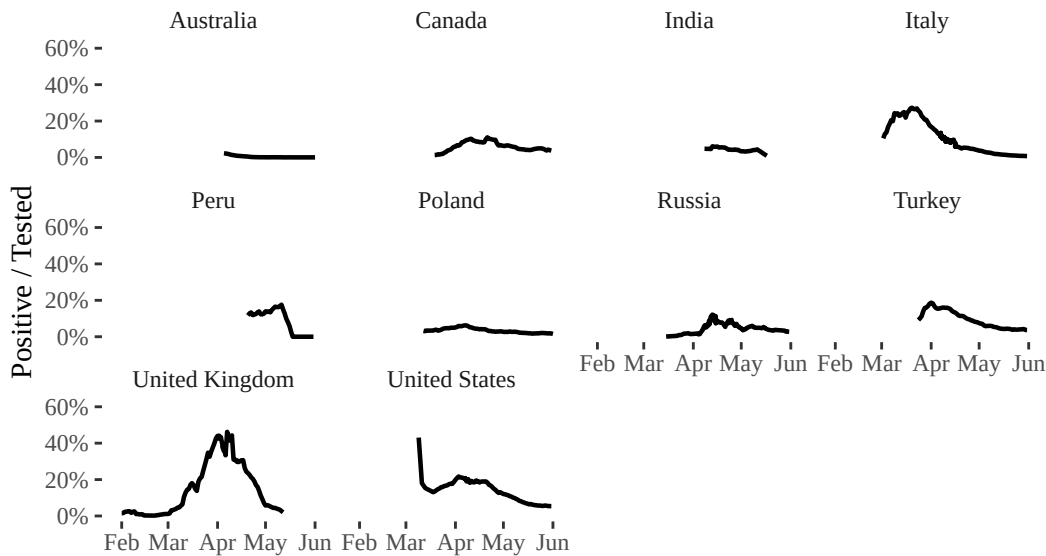
covid_ratio_time <- covid_df_all_states_daily %>%
  filter(Country_Region %in% countries) %>%
  mutate(Date = as.Date(Date),
         ratio = daily_positive / daily_tested) %>%
  filter(!is.na(ratio), is.finite(ratio), daily_tested > 0) %>%
  group_by(Country_Region) %>%
  arrange(Date) %>%
  mutate(ratio_7day = rollmean(ratio, k = 7, fill = NA, align = "right")) %>%
  ungroup()

ggplot(covid_ratio_time, aes(x = Date, y = ratio_7day)) +
  geom_line(linewidth = 0.8, na.rm = TRUE) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1),
                    limits = c(0, 0.6)) + # cap at 60% to remove UK spike
  labs(
    title = "7-Day Rolling Average: Positive-to-Tested Ratio",
    subtitle = "Top 10 countries by testing volume, January-June 2020",
    x = NULL,
    y = "Positive / Tested",
    colour = NULL
  ) +
  theme_tufte() +
  theme(legend.position = "bottom") +
  facet_wrap(~Country_Region)

```

7-Day Rolling Average: Positive-to-Tested Ratio

Top 10 countries by testing volume, January–June 2020



The time-series chart reveals several patterns obscured by the aggregate analysis.

The UK stands out as a data quality story. The spike to ~42% in mid-March then sudden drop is almost certainly not real. It reflects the UK's early testing strategy being restricted to hospitalised patients, making the ratio artificially high. When testing expanded, the ratio collapsed immediately.

The US and Turkey follow a similar arc. Both peak around late March to early April at ~20%, then decline steadily as testing capacity expanded — the classic pattern of a country catching up with an outbreak.

Italy peaks earliest and falls fastest, consistent with it being the first European country seriously hit and the first to scale testing in response.

Australia and Russia are the most interesting comparison. Australia stays near 0% throughout, reflecting aggressive early containment. Russia's later rise suggests the outbreak arrived later but testing did not keep pace.

Across all countries, ratios rise then converge toward low values by June. The aggregate positive-to-tested ratios reported in Section 6 therefore largely capture peak testing constraints rather than total disease burden, reinforcing the caveat raised in Section 2.

12 Putting It All Together

```
question_list <- list(
  "Which countries have had the highest number of deaths due to COVID-19?",
```

```

    "Which countries have had the highest number of positive cases against the
    number of tests?",
    "Which countries have made the best effort in terms of the number of COVID-19
    tests conducted related to their population?",
    "Which countries were ultimately the most and least affected related to their
    population?"
  )

answer_list <- list(
  "Death" = death_top_3,
  "Positive tested cases" =
  positive_tested_top_3,
  "The best effort in test related to the population" =
  best_effort_tested_cased_top_3,
  "The most affected country related to its population" =
  most_affected_country,
  "The least affected country related to its population" =
  least_affected_country
)

dataframe_list <- list(
  covid_df,
  covid_df_all_states,
  covid_df_all_states_cumulative,
  covid_df_all_states_daily
)

matrix_list <- list(covid_mat, covid_mat_rank)

vector_list <- list(vector_cols, population, countries)

data_structure_list <- list(
  "dataframe" = dataframe_list,
  "matrix" = matrix_list,
  "vector" = vector_list
)

```



```
covid_analysis_list <- list(question_list, answer_list, data_structure_list)
```

```
summary_table <- tibble(
  Question = c(
    "Highest deaths",
    "Highest positive-to-tested ratio",
    "Best testing effort per capita",
    "Most affected country (per capita)",
    "Least affected country (per capita)"
  ),
  Answer = c(
    paste(death_top_3, collapse = ", "),
    paste(names(positive_tested_top_3), collapse = ", "),
    paste(best_effort_tested_cased_top_3, collapse = ", "),
    most_affected_country,
    least_affected_country
  )
)
```

```
knitr::kable(summary_table, col.names = c("Question", "Answer"))
```

13 Summary

```
knitr::kable(summary_table, col.names = c("Question", "Answer"))
```

Question	Answer
Highest deaths	US, IT, UK
Highest positive-to-tested ratio	United Kingdom, United States, Turkey
Best testing effort per capita	United States, Russia, Italy
Most affected country (per capita)	Italy
Least affected country (per capita)	India

Answer:

- The United Kingdom, United States, and Turkey recorded the highest ratios of positive COVID-19 cases relative to tests conducted among the top ten most-tested countries in

the period studied, with ratios of approximately 0.11, 0.10, and 0.08 respectively. However, the time series analysis shows that those ratios are largely an artefact of testing strategy.

- The US had both the highest deaths and a high positive ratio, suggesting systemic undertesting. Italy had the second highest deaths and peaks earliest in the time series, supporting the interpretation that early undertesting drove high rates.

14 Technical Appendix: Concepts and Trade-offs

14.1 Named Vectors vs. Dataframes

The exercise uses named vectors as an intermediate step, extracting columns, naming them with country labels, and computing ratios element-wise. This is idiomatic base R but the tidyverse offers a more robust alternative.

14.1.1 The base R approach

`setNames()` is a concise way to assign names inline rather than as a separate step:

```
# These two are equivalent
eg_tested <- setNames(covid_top_10$tested, covid_top_10$Country_Region)

eg_tested <- covid_top_10$tested
names(eg_tested) <- covid_top_10$Country_Region
```

14.1.2 The tidyverse approach

Rather than extracting columns into separate vectors, the ratio can be computed directly on the dataframe using `mutate()`, keeping country names attached to the data throughout and avoiding several single-use intermediate objects. `deframe()` from the tibble package then converts a two-column dataframe into a named vector, with the first column as names and the second as values.

14.2 The `select()` Pattern: Positive vs. Negative Selection

Two complementary patterns appear in this analysis:

```
# Keep only these columns
covid_df_all_states_daily <- covid_df_all_states %>%
  select(all_of(daily_cols))

# Keep everything except these columns
covid_df_all_states_cumulative <- covid_df_all_states %>%
  select(-all_of(daily_cols))
```

The `-` prefix means “exclude”. Both use `all_of()`, the safe way to pass a character vector of column names into `select()` — without it, `tidyselect` raises a warning in recent versions of `dplyr`.

14.3 Why `filter()` Before `select(-Province_State)`

After filtering to `Province_State == "All States"`, every remaining row has the same value in that column, so it carries no information and is dropped. Chaining both steps in a single pipe is cleaner than `df$col <- NULL`, which mutates the object in place rather than producing a new one.

14.4 List Structure as a Data Container

Nested lists bundle heterogeneous outputs — strings, named vectors, dataframes, matrices — into a single object. In a larger project this might be returned from a function or saved with `saveRDS()`. Elements are accessed by position (`[[1]]`, `[[2]]`) or by name if the list is named.

14.5 `skim()` vs. `glimpse()` vs. `str()`

Function	Best for	PDF output
<code>str()</code>	Compact type and value preview, base R	Can overflow narrow pages
<code>glimpse()</code>	Tidyverse equivalent of <code>str()</code>	Wraps cleanly
<code>skim()</code>	Full summary statistics per column	Renders as a tidy table

For PDF reports, `glimpse()` is generally preferable to `str()` for structural inspection, and `skim()` is the best choice when you also want completeness rates and distributional summaries in a single call.